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The Assessment of Employees' Competences on the Labour Market in the Context of the Development of Artificial Intelligence Systems

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ABSTRACT

The rapid development of artificial intelligence has led to major changes in all areas of activity, particularly influencing the content of socio-economic activities and the profile of the labour market. In these circumstances, it is necessary to assess the competencies of employees in order to determine the extent to which organisations are acting and/or are prepared to meet the new challenges generated by the development of artificial intelligence systems. In this context, the present research aims to analyse the level of appreciation of employees' competencies from an organisational perspective, identifying three sets of competencies by employee category: workers, specialists, and managers. The competencies included in the research were adapted from the Continuing Professional Development Survey applied in organisations in EU Member States. The observation method used in the research was the survey, which included 350 organisations, and the respondents were HR managers or managers of organisations, depending on their size. In addition to the bibliometric analysis of the field, a set of statistical methods was applied in processing, analysing, and interpreting the respondents' answers: ANOVA analysis, association analysis, rank method and text analysis. The results show some differences in the assessment of the required competencies across different categories of employees, as well as according to the size and scope of the organisation. The results reflecting the opinion on the appreciation of employees' competencies are interpretable also according to the way each manager understands IT development as a promoter of the promotion of artificial intelligence. The results obtained show that IT skills were rated at the lowest level compared to other categories of skills. The research highlights those areas of work where some employee skills are valued at a lower level compared to other areas of work.

INTRODUCTION

Artificial intelligence (AI) has penetrated all spheres of life, including working life, leading to a shift in the work paradigm and the demands on employees' professional skills. Depending on the field of work and the content of the job, AI is being applied differently at the current stage, but with big implications for the future need for skills in human resources. The relationship and impact of AI on the labour market and skills needs have been of concern to several researchers (Acemoglu & Autor, 2011; Arup et al., 2023).

In addition, many researchers have analysed the implications of AI on different areas of activity. Thus, the impact of using AI in the following domains has been researched: retail (Purcărea et al., 2021; Stanciu & Rândaşu, 2021); finance and banking (Weber, 2023); industry (Borboni et al., 2023); agriculture (Ganeshkumar et al, 2023). In order to adapt to the new changes, organisations need, on the one hand, to adjust new work and organisational models (Jaiswal et al., 2022) and, on the other hand, to reassess the required skills of employees, as the automation of certain work tasks, which may lead to retraining or the development of new skills (Hancock et al., 2020).

Although AI is substantially changing the profile of activities and jobs, Autor (2015) believes that a significant category of medium-skilled jobs that combine specific job skills with literacy, numeracy, adaptability, problem-solving and common-sense skills will persist for decades to come. The researcher's argument is based on the idea that many of the future medium-skilled jobs will combine routine technical tasks with non-routine ones, where employees have comparative advantages: interpersonal interaction, flexibility, adaptability and problem-solving.

The rapid development of AI has led to a drastic increase in the demand for IT professionals, and the lack of skills in AI and IT, respectively, poses a significant challenge to the labour market, including academia (Bawack et al., 2021). Thus, both academia and business need to react quickly by designing and expanding academic and continuing education programmes in line with labour market needs to train new generations of IT and AI professionals (Zhang et al., 2022). Labour shortages have also been analysed in Romania, including in the IT&C sector (Chivu et al., 2019). The authors of the research argue that even though the IT&C sector has been declared a strategic priority by policy makers at the national level, with several measures being promoted to support the sector, the labour shortage is still high and has increasing trends. In the same vein, Bîrcă et al. (2023) analysed skills in the labour market in the Republic of Moldova.

Based on the above, this paper aims to determine how businesses (organizations) assess the competencies of their employees, especially in IT, in the scope of the development of AI systems. The orientation of this research is driven by the fact that economic agents need to be aware of the relevance of continuous training of employees' competencies for the achievement of organizational goals, but also of the need for investment in training new IT competencies for employees, as a result of changing work content and work processes. Moreover, the development and upgrading of employee skills can take place through the initiation of training programmes by organisations in the context of lifelong learning, building on the current skill level of employees in organisations. This directly involves the regular development of digital skills in the use of information technology as a tool for knowledge and lifelong learning.

This paper is structured in three sections. The first part reviews the literature on managers' assessment of employee competencies in their organisations from the perspective of IT development and artificial intelligence. In the second part, the applied research methodology is presented, highlighting the assessment of competencies included in the research, followed by the main results obtained. The paper concludes with the presentation of the main conclusions and managerial implications on the training and upgrading of employees' skills in the context of accelerated AI development.

1. LITERATURE REVIEW

Broadly, AI encompasses systems that are capable of mimicking intelligent human behaviour (Kok et al., 2009) and refers to the study and development of "intelligent agents" (machines, software or algorithms) that act intelligently by recognising and responding to their environment (Agrawal et al., 2018). In another approach, AI consists of systems that can process data and information, similar to intelligent

behaviour, including aspects related to reasoning, learning, perception, prediction, planning or control (UNESCO, 2021).

AI leads to the automation of certain processes and work tasks, allowing employees to focus more on tasks of greater complexity. According to Morandini et al. (2023), this can lead to the development of new skills and the optimization of existing skills, allowing employees to be more productive and innovative. While Morandini et al., (2023) view AI as a source of motivation for learning, Georgieff & Hyee (2022) view it as a substitute for the workplace.

OECD (2018) classifies competencies into: cognitive and meta-cognitive competencies (critical thinking, creative thinking, learning to learn and self-management), non-cognitive competencies (empathy and collaboration) and digital competencies (use of digital devices). Non-cognitive skills include: open-mindedness, openness to learning and change, flexibility, curiosity, innovation, creativity, entrepreneurship, resilience, planning, organisation, responsibility, persistence, teamwork, communication, initiative, sociability, empathy, collaboration, emotional control and positivity (Kautz et al. 2014). In turn, digital skills can be defined as a set of core knowledge, skills and other characteristics that enable people to perform their work tasks effectively in a digital environment (Oberländer et al. 2020).

The integration of AI systems in organisations has led to a growing awareness of the importance of identifying and cultivating cross-functional competencies among employees. Cross-cutting competencies, also known as transferable or soft competencies, are those that can be applied in different areas of work and for performing different work tasks (Hart et al., 2021). According to the authors, cross-cutting competencies are necessary or valuable for effective action in any type of work, learning or life. They are considered "transversal" because they are not exclusively linked to a specific context and are becoming increasingly important in a society experiencing rapid technological and social change. In this context, Hart et al. (2021) propose a model for transversal competencies consisting of six main categories: basic skills, thinking skills, self-management skills, communication and interpersonal skills, physical and manual skills and life skills. In the context of AI development, basic skills refer to an individual's ability to understand, speak, read and write in one or more languages, as well as being able to work with numbers and use digital devices and applications. Thinking skills show the individual's ability to apply mental processes to collect, conceptualise, analyse, summarise and/or evaluate information obtained or generated through observation, experience, reflection, reasoning or communication (Morandini, 2023). Self-management skills are reflected in the individual's ability to carry out activities in a reflective, responsible and structured manner according to certain values, accepting feedback and identifying opportunities for personal and professional development (Hart et al., 2021). Communication and interpersonal skills refer to an individual's ability to interact positively and productively with others and can be demonstrated by: communicating ideas effectively and empathetically, aligning one's own goals and actions with those of others, seeking solutions to disagreements, building trust and resolving conflict, caring for the well-being and progress of others, managing activities through appropriate leadership (Morandini, 2023). Physical and manual skills show the individual's ability to perform tasks and activities that require manual dexterity, agility and/or physical strength. They can be performed in difficult or hazardous work environments requiring stamina or strength. (Morandini, 2023) According to Wamba-Taguimdje et al. (2020), improving physical and manual skills increases adaptability to new technologies and work environments, as workers skilled in the use of different types of machines or equipment can more easily adapt to new AI technologies that are introduced in the workplace. Life skills are skills that model adaptive and positive behaviour, enabling individuals to cope effectively with daily demands and challenges (Haug et al., 2021).

2. RESEARCH METHODOLOGY

In order to assess the level of skills among employees in organisations in the Republic of Moldova, we used the Continuing Professional Development Survey in organisations, applied in EU Member States and carried out every five years. It should be noted that in the Republic of Moldova this survey is not applied in domestic organizations. Also, the National Bureau of Statistics of the Republic of Moldova does not apply a survey that collects data on continuing vocational training and employees' skills, except for questions on vocational training included in the Labour Force Survey.

The Survey of Continuing Vocational Training in Organisations, applied at EU Member State level, includes 11 categories of competencies (Table 1).

Table 1. Professional competencies according to the European Commission

No.	*Competencies*	*Description of competence*
1.	General IT skills.	Use of computers, word processing, electronic diary, simple spreadsheets or internet.
2.	IT professional skills.	Specialist knowledge or understanding, such as producing web pages and developing complex programs.
3.	Management skills	Managing staff and planning the activities of others.
4.	Team working skills	Relations with colleagues, cooperation in work.
5.	Working with clients	Customer relations, persuading or influencing others.
6.	Problem solving	Identifying problems or shortcomings, establishing causes and considering solutions.
7.	Office administration skills	Billing, time management.
8.	Foreign language competencies	Reading, writing, listening and speaking in a foreign language.
9.	Job-specific competencies	Operation of machinery and plant; Sale of products or services.
10.	Oral and written communication	Making speeches or presentations. Reading long documents such as reports, manuals, articles or books.
11.	Numeracy and/or literacy competencies.	Simple arithmetic, calculations using decimals, percentages or fractions. Reading or writing information such as forms, notices, signs or short documents.

Source: European Commission EUROSTAT (2022).

Based on the competencies formulated by the European Commission (2022), they have been adapted to the situation in the Republic of Moldova, establishing sets of competencies by categories of human resources: workers, specialists and managers. Thus, a set of 8 competencies was established for workers, and 10 and 11 competencies - for specialists and managers (Table 2).

Table 2. Competences covered in the research by categories of employees

No.	*Workers*	*Specialists*	*Managers*
1.	Literacy skills	Literacy skills	Literacy skills
2.	Numeracy skills	Numeracy skills	Numeracy skills
3.	Oral and written communication skills	Oral and written communication skills	Oral and written communication skills
4.	Foreign language skills	Foreign language skills	Foreign language skills
5.	General IT skills	General IT skills	General IT skills
6.	Professional IT skills	Professional IT skills	Professional IT skills
7.	Job-specific skills	Job-specific skills	Job-specific skills
8.	Team working skills	Team working skills	Team working skills
9.		Interpersonal skills	Interpersonal skills
10.		Intercultural skills	Intercultural skills
11.			Management skills
Cronbach's Alpha	*0.880*	*0.917*	*0.939*

Source: Processed by authors in SPSS

In our research we have included intercultural competence for specialists and managers, taking into account the process of globalization of the labour market but also from the perspective of the penetration

of the internal market by labour force from outside the Republic of Moldova. In order to assess each competency by employee category, a questionnaire was developed that included the job competencies shown in Table 2. Respondents were asked to rate the competencies for each employee category on a Likert scale from "1" to "5", where "1" was the lowest rating and "5" - the highest rating. The respondents were HR managers - in the case of large and medium-sized enterprises, and in the case of small enterprises - their managers or HR managers. The data collection process took place online by filling in the questionnaire in Google Forms.

The collective for determining the sample was formed by economic agents from the Republic of Moldova. The sample consisted of 350 respondents (employers). The structure of respondents according to the number of employees is as follows: 62 respondents - micro enterprises (up to 9 employees), 100 respondents - enterprises with 10-49 employees, 105 respondents were part of the medium-sized enterprises category, and 83 respondents represented large enterprises. Table 3 also shows the structure of respondents according to a set of characteristics of organisations: size of organisation, location, type of ownership and field of activity.

Table 3. Profile of the sample of respondents by characteristics of organisations

Feature	*Categories*	*n= 350*	*%*
Size of the organisation	Up to 9 employees	62	17,7
	10- 49 employees	100	28,6
	50 - 249 employees	105	30,0
	Over 250 employees	83	23,7
Field of activity	Agriculture and forestry	41	11,7
	Trade	69	19,7
	Services	48	13,7
	Industry	56	16,0
	Hotels and restaurants	21	6,0
	Health and social care	13	3,7
	Transport and communications	23	6,6
	Financial activities	41	11,7
	IT	10	2,9
	Construction	16	4,6
	Other	12	3,4

Source: Processed by authors in SPSS

3. RESULTS AND DISCUSSION

Starting from the research objective of identifying the degree of appreciation of skills in the labour market in the view of employers in the context of the development of artificial intelligence systems, the authors set out to identify the level of appreciation of general and professional IT skills by top organizational management compared to other skills. From Figure 1, it can be easily observed that both at the level of workers, specialists and management, the level of appreciation of having general and professional IT skills is below the level of other skills. This suggests that decision-makers are aware of the need to develop IT skills in the context of the development of artificial intelligence systems.

For the in-depth analysis that we carried out further, we excluded the opinion of respondents from micro-enterprises in order to highlight the specific profile of larger organizations regarding the skills of employees in the context of AI development. The assessment of employee skill levels was carried out by statistical analysis using the ANOVA procedure (Jaba, 2002). For each employee category, the skill levels of managers, specialists and workers were compared between different categories of enterprises defined by the size of the organisation and their field of activity.

	Literacy skills	Numeracy skills	Oral and written communication skills	Foregn language skills	General IT skills	Professional IT skills	Job-specific skills	Team working skills	Relationship skills	Cross cultural skills	Management skills
Worker	3,8	3,6	3,8	2,8	2,7	2,5	4	4,2			
Specialist	4,5	4,5	4,6	4	4,1	3,7	4,5	4,5	4,4	4,2	
Manager	4,6	4,6	4,7	4,3	4,3	4	4,6	4,6	4,6	4,6	4,5

Figure 1. Competences assessment by categories of employees

Source: Processed by authors in SPSS

For managers, the skill level of this category of employees is higher in larger firms (Table 4). For seven of the 11 skills assessed, higher scores for managers are recorded for firms with more than 250 employees. Thus, there is a linear relationship between the level of competencies and firm size. However, managers in companies with 10-49 employees have higher skill levels in language skills, professional IT skills, teamwork and interpersonal skills. These differences observed by company category are not statistically significant (the probability associated with the Fisher test is above the chosen significance threshold of 10%, 5% and 1% respectively).

The competence level of specialists is higher in enterprises with more than 50 employees than in small enterprises with 10-49 employees (Table 4). For all ten competencies, the probability associated with the Fisher test is above the chosen significance threshold of 10%, 5% and 1%, respectively. For workers, higher skill levels are observed overall in enterprises with 10-49 employees and in those with more than 250 employees compared to medium-sized enterprises (Table 4). Thus, looking at the oral and written communication skills as well as the language skills of workers, the Fisher test indicates, at 10% and 5% risk respectively, a significantly lower level of skill appreciation for workers in medium-sized enterprises compared to the other two categories of enterprises. Similarly, in the case of job-specific skills, the Fisher test indicates a significantly higher level of competence among workers in large enterprises compared to medium-sized enterprises at 10% risk.

Table 4. Output of means and Fisher's test on the assessment of employees' skills by categories of firms differentiated by number of employees

Skills	*Company size*	*Manager*		*Specialist*		*Worker*	
		Mean	*Fisher test*	*Mean*	*Fisher test*	*Mean*	*Fisher test*
Literacy skills	10 - 49 employees	4.66	0.250	4.55	1.036	3.92	1.254
	50 - 249 employees	4.55		4.72		3.71	
	Over 250 employees	4.72		4.60		3.96	
Numeracy skills	10 - 49 employees	4.63	1.082	4.47	0.973	3.77	2.317
	50 - 249 employees	4.47		4.63		3.47	
	Over 250 employees	4.63		4.60		3.75	
Oral and written communication skills	10 - 49 employees	4.57	1.065	4.56	1.025	3.91	2.854*
	50 - 249 employees	4.56		4.64		3.63	
	Over 250 employees	4.64		4.72		3.95	
Foreign language skills	10 - 49 employees	4.64	1.932	3.98	1.819	3.04	4.136**
	50 - 249 employees	3.98		3.91		2.49	
	Over 250 employees	3.91		4.23		2.83	
General IT skills	10 - 49 employees	4.03	1.100	4.06	1.326	2.73	.978
	50 - 249 employees	4.06		4.16		2.50	
	Over 250 employees	4.16		4.31		2.75	
Professional IT skills	10 - 49 employees	4.17	1.444	3.70	.084	2.55	1.408
	50 - 249 employees	3.70		3.76		2.28	
	Over 250 employees	3.76		3.70		2.24	
Job specific skills	10 - 49 employees	3.72	2.049	4.49	1.697	4.04	2.402*
	50 - 249 employees	4.49		4.57		3.91	
	Over 250 employees	4.57		4.71		4.25	
Team working skills	10 - 49 employees	4.58	1.420	4.48	.354	4.25	1.063
	50 - 249 employees	4.48		4.57		4.06	
	Over 250 employees	4.57		4.57		4.23	
Interpersonal skills	10 - 49 employees	4.54	1.922	4.39	.909		
	50 - 249 employees	4.39		4.51			
	Over 250 employees	4.51		4.54			
Intercultural skills	10 - 49 employees	4.48	.391	4.28	.070		
	50 - 249 employees	4.54		4.26			
	Over 250 employees	4.59		4.31			
Management skills	10 - 49 employees	4.48	1.297				
	50 - 249 employees	4.61					
	Over 250 employees	4.75					

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Source: Processed by authors in SPSS.

Comparative analysis of skills by employee category and by field of activity reveals significant differences in IT (professional) skills for managers (Table 5). Thus, for an assumed risk of 10%, hotel and restaurant managers have a significantly lower level of IT (professional) skills than managers in Commerce, Services, Industry, Transport and Communication, Financial Activities and IT. Managers in enterprises other than the 10 sectors covered by the study also have significantly higher IT (professional) skills for an assumed risk of 10% than managers in Agriculture and Forestry, Trade, Services, Industry, Hotels and restaurants, Health and social work and Construction.

Table 5. The output of means and Fisher's test on the assessment of the level of employees' skills according to the field of activity of the firms

Domain / *Skills*	*Agriculture and forestry*	*Trade*	*Services*	*Industry*	*Hotels and restaurants*	*Health and social care*	*Transport and communica-*	*Financial activities*	*IT*	*Construction*	*Other*	*Fisher test*
Manager												
Literacy skills	4.59	4.68	4.96	4.75	4.77	4.82	4.33	4.60	4.78	4.79	4.85	0.772
Numeracy skills	4.62	4.68	4.92	4.77	4.46	4.45	4.44	4.58	4.78	4.57	4.77	0.733
Oral and written communication	4.76	4.71	4.92	4.66	4.54	4.91	4.56	4.83	4.89	4.71	4.69	0.617
Foreign language skills	4.24	4.36	4.25	4.30	3.77	4.45	4.44	4.58	4.56	4.21	4.92	1.233
General IT skills	4.29	4.46	4.54	4.25	3.85	4.09	4.22	4.45	4.56	4.50	4.92	1.249
Professional IT skills	3.88	4.14	4.00	3.96	3.23	3.82	4.17	4.18	4.67	3.57	4.77	1.774*
Job specific skills	4.65	4.63	4.75	4.71	4.54	4.73	4.50	4.65	4.78	4.57	4.77	0.239
Team working skills	4.68	4.61	4.75	4.70	4.77	4.82	4.56	4.63	4.78	4.71	4.92	0.366
Interpersonal skills	4.59	4.63	4.83	4.68	4.46	4.82	4.56	4.63	4.67	4.50	4.92	0.610
Intercultural skills	4.35	4.52	4.54	4.54	4.46	4.36	4.61	4.55	4.78	4.64	4.85	0.494
Management skills	4.62	4.54	4.79	4.57	4.62	4.55	4.61	4.68	4.67	4.71	4.92	0.413
Specialist												
Literacy skills	4.56	4.66	4.96	4.71	4.62	4.82	4.17	4.50	4.89	4.64	4.38	1.224
Numeracy skills	4.44	4.55	4.92	4.64	4.23	4.09	4.50	4.55	4.78	4.71	4.54	1.163
Oral and written communication skills	4.62	4.59	4.92	4.59	4.62	5.00	4.39	4.65	4.78	4.57	4.54	0.860
Foreign language skills	3.74	3.93	4.25	3.80	4.15	4.00	4.17	4.50	4.44	3.64	4.15	1.552
General IT skills	4.00	4.09	4.42	4.18	4.23	4.00	4.17	4.28	4.44	3.79	4.46	0.669
Professional IT skills	3.65	3.75	3.71	3.52	3.38	3.91	3.89	3.93	4.33	3.36	4.00	0.824
Job specific skills	4.62	4.48	4.58	4.68	4.62	4.91	4.56	4.53	4.56	4.57	4.46	0.392
Team working skills	4.68	4.45	4.67	4.50	4.62	4.82	4.50	4.45	4.56	4.57	4.46	0.398
Interpersonal skills	4.59	4.41	4.58	4.43	4.31	4.55	4.56	4.48	4.67	4.36	4.54	0.308
Intercultural skills	4.29	4.27	4.50	4.09	4.46	4.18	4.50	4.30	4.78	4.00	4.23	0.763
Worker												
Literacy skills	3.65	3.82	4.08	3.91	4.08	3.73	3.67	3.80	4.33	3.57	4.23	0.679
Numeracy skills	3.56	3.75	3.67	3.57	3.62	3.55	3.56	3.83	3.78	3.43	3.69	0.278
Oral and written communication skills	3.85	3.68	4.13	3.68	3.85	4.27	3.56	3.88	3.67	3.50	4.62	1.706*
Foreign language skills	2.35	2.77	3.00	2.36	2.92	2.73	2.94	3.58	2.56	2.21	3.31	2.782***
General IT skills	2.24	2.66	2.75	2.27	2.69	2.82	2.50	3.33	2.67	2.43	3.38	2.303**

Professional IT skills	2.09	2.30	2.50	1.77	2.15	2.82	2.56	2.95	2.56	2.29	3.31	2.833***
Job specific skills	3.94	4.02	4.25	4.14	3.92	4.00	3.89	4.15	3.89	3.93	4.15	0.310
Team working skills	4.44	4.05	4.38	4.07	4.23	4.00	4.06	4.13	3.56	4.29	4.77	1.306

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Source: Processed by authors in SPSS

In the case of specialists, the level of skills does not differ statistically significantly according to the field of activity of the companies (the probability associated with the Fisher test is higher than the chosen significance threshold of 10%).

For workers, there are significant differences by occupation in language skills, IT (professional) skills, general IT skills and communication skills (Table 5). Employees in organisations performing Financial Activities have a significantly higher level of general IT skills compared to employees in Agriculture and Forestry, Trade, Services, Industry, Transport and Communication and Construction. Workers in other industries also have a higher level of general IT skills than workers in Agriculture and Forestry, Trade, Industry, Transport and Communications and Construction. Workers in financial organisations have significantly higher levels of IT (professional) skills than those in Agriculture and Forestry, Trade, Industry and Hotels and Restaurants. At the same time, workers in other industries have higher IT (professional) skills than workers in Agriculture and Forestry, Trade, Industry, Hotels and Restaurants and Construction. The level of language skills is significantly higher among workers in organisations carrying out financial activities compared to those in Agriculture and Forestry, Commerce, Hotels and Restaurants, Healthcare, IT and Construction. In terms of written and oral communication skills, a significantly higher level of skills is identified for workers in Services compared to those in Commerce, Industry, Transport and Communication and Construction. Workers in Health and Social Care also have a higher level of communication skills than those in Trade, Transport and Communications and Construction.

In addition to assessing the competencies of employees, at the level of the companies observed, we aimed to identify the professional competencies that employees in the respondent organisations would like to develop. In this case, the research also included the responses provided by microenterprises to see what expectations they have in terms of professional skills. In this context, we started from Gallardo-Gallardo & Collings' (2021) statement that organisations should rely on competencies rather than job descriptions or traditional roles.

In this vein, from the previously assessed competencies, *"Literacy skills", "Numeracy skills", "Interpersonal skills"* and *"Intercultural skills" were* excluded, and for logical reasons, 4 other competencies were introduced, namely: "*Customer handing skills", "Time management skills", "New idea generation skills"* and *"Problem solving skills"* (Table 6).

From the analysis of the ranks of the professional competencies that should be developed in the employees of the studied organizations, they show different positions depending on the size of the organization (Table 6). The need to develop general IT skills is more evident in companies with 10-49 employees than in other companies.

Table 6. Hierarchy of professional competencies that should be developed in the employees of the studied organisations (rank method)

Size of the organisation / *Skills*	*The row*				
	Total	*Up to 9 employees*	*10- 49 employees*	*50 - 249 employees*	*Over 250 employees*
1. Oral and written communication skills	11	7	11	11	11
2. Foreign language skills	8	5	8	8	8
3. General IT skills	9	11	7	8	8
4. Professional IT skills	10	10	10	10	10
5. Job specific skills	1	2	3	1	3
6. Team working skills	3	3	2	3	1
7. Management skills	5	8	5	5	4
8. Customer handing skills	2	1	1	3	5
9. Time management skills	7	4	8	7	7
10. New idea generation skills	6	9	6	6	6
11. Problem solving skills	4	6	4	2	2

Source: Processed by authors.

From the set of skills analysed, by applying the association analysis and the χ test[2] (Jaba, 2002), a statistically significant association was obtained between the skills acquired and those required only for language skills (knowledge of a foreign language). Thus, respondents in organisations where workers have a higher level of language skills believe that these skills should be developed at the organisational level (χ^2 (4)= 10.931, p=0.027). At the same time, in the case of specialists, employers in organisations where there are specialists with a very high level of language skills consider these skills to be important for the organisation, while employers in organisations where specialists have no language skills or where the level of these skills is not very high consider these skills to be unimportant (χ^2 (4)= 9.073, p=0.059). Similarly, in organisations where managers have a very high level of language skills, there is a desire to develop these skills, whereas in organisations where managers have a low or medium level of skills, there is no desire to develop these skills (χ^2 (4)= 11.016, p=0.026).

For the other competencies studied, there was no significant association between the level of employee competencies and the organisation's intention to develop or not develop that competency.

CONCLUSION

This paper sought to highlight employers' perceptions of the competencies of employees within organisations in the context of the accelerating development of AI systems, and how employers relate to the different competencies needed in the future to achieve organisational goals. In this context, a set of employee competencies, considered relevant for the effective work of organisations, were assessed, but also in the perspective of the introduction of new technologies that imply the need to adjust current employee competencies.

The results obtained show that IT skills were rated at the lowest level compared to other categories of skills, which allows us to conclude that managers of organizations, regardless of their size and field of activity, have certain fears about the level of training of employees in the IT field and are increasingly aware of the relevance of this skill in the performance of job-related tasks. Similarly, language skills (communication in a foreign language) are valued at a lower level by respondents. This proves that in a globalised society, knowledge of a foreign language is imperative for the smooth running of business, regardless of the size or field of activity of the organisation.

Our research also highlights those areas of work where some employee skills are valued at a lower level compared to other areas of work, which could lead managers of these organisations to take concrete

steps in the future to help improve current skill levels and adapt them to the rigours dictated by the evolution of IT and AI systems.

Even if some authors argue that in the context of AI development, employee training should be oriented towards the development of soft skills (Hart et al, 2021; Morandini, 2023), the research results show that managers of domestic organizations are more inclined to develop technical, job-specific skills (57.3%), skills that are devaluing faster over time due to technological developments. Of the soft skills that managers of domestic organisations would most like to develop for their employees, the most important are those related to teamwork, working with customers and problem solving.

The original contribution of the paper lies in the fact that the aim was to evaluate professional competencies by categories of employees - workers, specialists and managers - in order to analyse their competencies in a differentiated way, which would allow senior managers in organisations to take decisions on the initiation and implementation of different training programmes for different categories of employees, oriented towards the development of certain competencies, based on their current development context.

From the perspective of involving managers in the process of developing the competencies of their organization's employees, our research provides them with the opportunity to objectively assess the level of competencies and make effective decisions related to continuing education programs that lead to increasing the value of employees' competencies. Likewise, the conducted study directs, to a certain extent, managers towards the need for soft skills development which becomes more relevant in the context of AI systems development.

This paper also has certain limitations. Firstly, it is a first attempt to assess the level of employee skills on the part of employers, which is why the results obtained cannot be compared with previous research showing the evolution of employee skills levels. Secondly, the level of representativeness of the respondent organisations does not fully correspond to the structure of the organisations by business area, which could influence to some extent the level of employee skills overall. Thirdly, the results reflecting employees' competencies are also interpretable according to each respondent manager's understanding of AI. Under these circumstances, respondents with different levels of knowledge in the field of IT and AI systems might have interpreted the level of employee skills differently.

As AI systems evolve, we aim to repeat this study to assess how employers' views of employee skill levels have changed. In the same way, we will try to analyse the extent to which AI systems have changed employers' perceptions of certain skills and, in particular, their assessment of the level of skills for different categories of employees.

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